

Jaiveer Bassi

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Experience

Physical Intelligence

May 2026 – Present

AI Robot Operator

San Francisco, CA

- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through diverse manipulation, sorting, and assembly tasks with precision and consistency.
- Perform quality control on AI-driven robot performance by evaluating model outputs against benchmarks, identifying failure modes and edge cases, and annotating task execution data to meet training-quality thresholds for model development.

House of Seva

June 2026 – Present

Founder & Executive Director

Vallejo, CA

- Founded a California nonprofit serving the South Asian community of Solano County, launching Seva Scholars, a college prep and mentorship program for first-generation youth.
- Designed, built, and deployed the organization's website (houseofseva.org) on Vercel with a serverless contact API (Resend), authenticated email (SPF/DKIM/DMARC), embedded donation flow, JSON-LD SEO, and PWA support.

Handshake AI

March 2026 – June 2026

Contracted AI Response Evaluator

Remote

- Evaluated LLM-generated code and technical responses across AI/ML and software engineering domains (C#/.NET, Python, ML systems), scoring on correctness, instruction following, and writing quality.

Kaiser Permanente Regional Laboratory

Oct 2023 – June 2024

Software Developer

Berkeley, CA

- Developed full-stack web application using React, Express, RESTful APIs, and SQL for uploading, processing, and viewing clinical lab instrument data across multiple departments.
- Built secure data pipelines with input validation and error logging for laboratory information systems, consolidating a manual data entry process across 50+ instrument data columns into an automated upload workflow.
- Collaborated directly with lab technicians and clinical staff to gather requirements, conduct demos, and iterate on features based on end-user feedback.

Projects

Habitua – iOS Tinnitus Therapy App | Swift, SwiftUI, AVAudioEngine, Swift Charts

- Built an iOS app from scratch with real-time audio synthesis using custom biquad IIR filters, pink/brown noise DSP, and a personalized notch filter tuned to the user's matched tinnitus pitch, with zero third-party dependencies.
- Designed guided habituation exercises and a progressive silence-exposure ladder grounded in Tinnitus Retraining Therapy principles, with Swift Charts tracking distress and awareness over time.

MedStract.net – Biomedical Research Tool | Python, Flask, PubMed API, NLP, JavaScript, Chart.js, ReportLab, Vercel

- Built a full-stack biomedical research tool with a Flask REST API that queries PubMed's 36M+ article database, with rate limiting and caching for reliable performance.
- Designed a custom NLP pipeline that rewrites research summaries for different audience levels, with PDF export, citation generation, and publication trend visualizations via Chart.js.

Castline Studio Order System – E-Commerce Order Platform | Next.js 14, TypeScript, WebAssembly, Vercel, REST APIs

- Built an order management web app in Next.js 14 (App Router, TypeScript) featuring real-time STL file analysis via a custom WebAssembly/JavaScript parser to calculate volume, dimensions, and print cost entirely client-side.
- Powers a \$2K+/month e-commerce business with Etsy REST API integration, automated shipping/email workflows, and a fully automated order pipeline from purchase to fulfillment.

CareerTuner.org – AI Resume Analyzer | OpenAI GPT-4, React, Node.js, Supabase, Tailwind, PDFKit

- Built and deployed an AI-powered resume analyzer integrating the OpenAI API for intelligent feedback, ATS scoring, and skills gap analysis.
- Implemented a RAG pipeline for context-aware resume suggestions tailored to specific job descriptions, with a Supabase backend and PDF export via PDFKit.

Research Publications

Brain Tumor Classification with Pretrained CNNs in PyTorch 7,000+ MRI, ResNet-18 DOI
Efficient Packaging and Deployment of AI Models for Edge Inference DOI
Transferability of Adversarial Attacks Across ML Models DOI

Technical Skills

Languages: Python, TypeScript, JavaScript, Swift, SQL, C++, Java, HTML/CSS
Frontend: React, Next.js, SwiftUI, Tailwind CSS, Bootstrap
Backend: Node.js, Express, Flask, RESTful APIs, Docker, Git, CI/CD, Serverless
AI/ML: OpenAI API, LangChain, PyTorch, HuggingFace, RAG Pipelines
Data & Cloud: PostgreSQL, MongoDB, Supabase, Vercel, AWS, Google Cloud, DICOM, HL7/FHIR
Other: WebAssembly, OAuth 2.0, DSP/Audio Signal Processing, Pandas, NumPy, OpenCV

Education

Grand Canyon University	May 2024 – Oct 2025
MS in Computer Software Engineering	Phoenix, AZ
University of Silicon Valley	May 2021 – Aug 2023
BS in Computer Science	San Jose, CA

Certifications

AWS Solutions Architect Associate (Apr 2026)	HackerRank Software Engineer Certificate (Mar 2026)
Cisco Network Support and Security (Sep 2025)	IBM Artificial Intelligence (Aug 2025)
Stanford AI/ML in Healthcare (Jun 2025)	Google Project Management (Sep 2024)